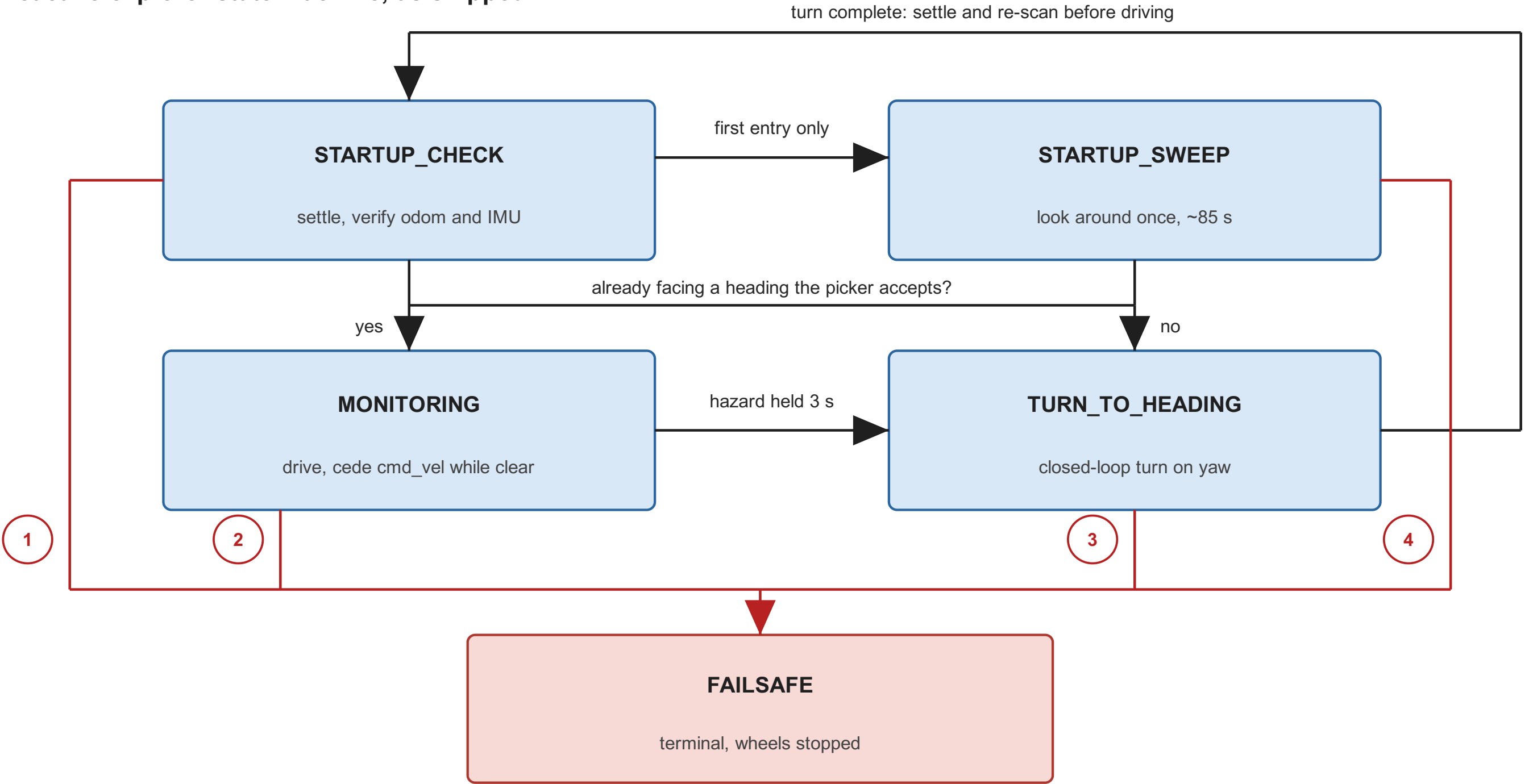


Reactive-explorer state machine, as shipped



IMU over-tilt preempts every state except STARTUP_CHECK

The rover stops, re-scans, and turns towards a clear heading. It never commands a manoeuvre of its own, and there is no reverse state.

A stale LiDAR scan (lidar_stale) ends the mission from any state.

Red transitions end the mission:

1 **STARTUP_CHECK**: imu_not_ready, over_tilted, boxed_in, terrain_confirm_timeout, terrain_rejected_everywhere, terrain_search_timeout

2 **MONITORING**: boxed_in, no_room_to_turn 3 **TURN_TO_HEADING**: turn_timeout, odom_stale

4 **STARTUP_SWEEP**: sweep_turn_timeout, no_room_to_turn, boxed_in